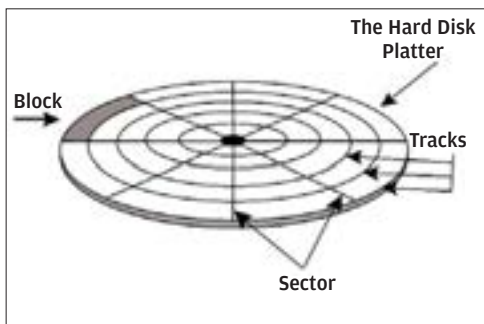


tors. Tracks are concentric circles that start at the centre of the disk and move outward to the edge. Blocks are the areas that lie between sectors and tracks: it is the blocks where data is stored.



The organisation of tracks, sectors and blocks on a disk

As users, we can't really be expected to understand how to store our data in these sectors and blocks, so our operating system gives us a *filesystem*, which is its way of protecting us from having to understand storage devices—by showing us our data in a friendlier way. To be more specific, the OS shows us our data organised into *directories*, which contain our files. Directories may also contain sub-directories, which could contain more sub-directories and files, and so on. While nearly all OSes organise data into directories, the ways they finally write the data to the disk are different.

The simplest filesystem is the *File Allocation Table* (FAT), which is understood by most OSes today. Let's look at it this way—the hard drive is a massive warehouse with many shelves (sectors), and each shelf with many boxes (blocks). The OS is the supervisor of this warehouse. Let's say that the warehouse is empty right now. A new truckload of files has arrived, and the OS must now put them away into the shelves. It starts at the first shelf and begins putting the files in their boxes. If a file is too big for the box, it splits it up and starts filling new boxes till the entire file is stored. While it stores the files, it takes notes in its FAT, which is its handy guide to all the files in the warehouse. The FAT tells it which file is stored in which shelf, and this helps it locate it quickly if a program asks for that file.